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ArchmageAU
HF Support



My HF Map location
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From tail-in to all 8s and funnels in 6 months

This is quite large. YOU HAVE BEEN WARNED.

From tail-in to all 8s and funnels in 6 months.

This is a summary of the thread "Just confirming I am on the right track" (https://www.helifreak.com/showthread.php?t=493289

As the title suggests. This is to take an absolute novice from tail-in hover to all orientation flying in less than 6 months.

I must thank those on HeliFreak that have helped me enormously along the way:
- gitbse, RogerRabit, clicky, LN400, blade450
And those who have taken the read and respond to my rantings on HeliFreak
- anthony1, AcidDrink, rooni79, Smok1n, mdbsat, lamk, Steve Graham, Stubbz, Ozzy Mozzy, HelisRock

An extra thank you to Code3Medic for converting my Heli-X recording sessions to video.

First a warning: This is not for everyone. It takes discipline, patience and is at times **very boring**. However, if wanting to fly RC he well, the results are worth the effort.

I progressed from tail-in to funnels (and more) in 6 months. When following this guide, you may take more time, you may take le Everyone progresses at their own rate. Do not feel discouraged if it takes longer than anticipated. There is no time limit. One thir is certain, more consistent and structured practice leads to faster progression.

What you will need:

- A good sim.
  - Preferably with a hover training mode.
  - Preferably with the ability to slow time.
  - Ideally uses your own transmitter (RealFlight's controller is fine though)
- A timer (I used my phone)
- A note taking method (I used a spreadsheet)
- A portable music player (I used my phone again)
- A collective pitch heli (To practice IRL what you learn. I used Blade micros (nano and 130x) because they are resilient (cras without breaking), and spares are quick and easy to obtain).
  - If learning on a larger heli (450 or greater), a FBL with a self levelling facility is highly recommended.

A note on sims. They can be very boring. The sound is often poor or distracting. Depth perception is an issue. But they are the Bl tool for learning quickly and safely when used as a tool. The portable music player is to help cut that boredom of some of the training sessions.

This method of learning relies HEAVILY on the proper use of the sim (as a training tool). I have read may threads where people complain they dislike using a sim. If you want to progress QUICKLY, the sim is really the only way to go as: The heli is always balanced, You never run out of fuel or battery, There are no crash costs, No wrenching to fix a broken heli, You cannot injure your or others, There is no restriction on space, No weather restrictions and No daylight restrictions.

The premise for the training is to make all orientation corrections automatic. After 6 months, I no longer think of how to correct i

any orientation. Most often, it just happens. I still occasionally get it wrong, but it's only 1% of the time (and I often get enough ti to re-correct).

So following this guide will be training an automatic feedback based motor-neuron skill. The ONLY way to do this is practice (an lots of it). You are trying to get your RC heli control skills to be like walking (assuming you have no limitations that prevent you walking). Walking is a VERY complex skill, but most of us do it so automatically it takes no thought at all. We got to that point because we did A LOT of it. We made lots of mistakes early on, learned from the mistakes and continued on. In learning progression it is at a state of "Unconscious competence".

Skills progression go:

- "Unconscious incompetence" (we don't know how to do it, and need guidance to begin)
- "Concious incompetence" (we know how to do it, but have little experience and often make mistakes)
- "Concious competence" (we can do it, but have to think how to do it each time, few mistakes)
- "Unconscious competence" (we can do it without thinking, rarely mistakes, can get faster and faster at it)

There are a number of skills that make up RC heli piloting. Each one needs to go through this progression sequence. Some will move faster than others. You should reach "upright tail-in hover" unconscious competence quickly, but then there is "upright nos hover", "upright side-on hover", "inverted hover", "circles", etc.. Reaching higher levels of competence in one will help raise skills i another. There are also skills in heli set-up and balancing, transmitter programming, flight physics, etc.. that help RC heli pilots, b this guide is not about those areas.

This guide assumes you can already hover tail-in, in normal mode. If not up to that, see RADD's school of rotary flight (see: RaddsFieldManual.pdf - attached now as the dream-models site where the original was housed is now offline). Even if you cann tail-in hover, the initial exercises should still work and will teach and improve that skill.

Fear:

This learning method should also help train you to be less fearful of real life flying. It will not completely eliminate this fear thou

Fear tends to be from a number of sources. In its base form, fear is a feeling of situational threat to which the subject does not f they have an adequate level of control over. This threat can be social (look foolish), financial (cannot afford a crash), personal (physical danger).

This training (heavily based on sims) helps increase a pilots level of control so that the threat is less imposing. By using a sim, there is no social, financial or physical threat. By learning how to be in total automatic control of the heli, know when the heli is o of control and how to bail-out safely, this reduces fear. Combine this training with a good auto-levelling system (as a last resort emergency save), and the pilot can be confident flying and landing their expensive heli, regardless of what happens. This reduce fear.

Transmitter handling:

I use and recommend pinch setting. It is not for everyone, quite a few top pilots use just thumbs. I found pinch helps my precisio follow the principles in the "1st U.S. R/C Flight School - Transmitter Handling" (http://www.rcflightschool.com/TransmitterHandling.pdf). I use pinch without a tray or neck strap, but I have long slender finger (can clear an octave +1 stretch on a piano).

Transmitter mode:

I use and recommend mode 2. I live in a country that is predominantly mode 1, but I chose mode 2 as I had a potential instructo that used mode 2. This method works with either, but most new pilots will progress faster using mode 2. In mode 2, all the cyclic controls (forwards/backwards/left/right) are on the right stick. Most internet instruction videos showing sticks are also mode 2. Most HeliFreakers are in the US and use mode 2.

Sim setup:

Whatever you do in the sim, you do in real life. Your fingers do not know the difference. Ideally your sim transmitter (TX) should have a throttle hold switch in the same place as your real TX (I use my flying TX as my sim TX).

My preferred sim for this training in Heli-X. It works in Windows, Mac and Linux as it is Java based. By using a TX to USB connection (refer Heli-X site for details - http://www.heli-x.net/support_e.shtml), you can use your TX. Heli-X's hover mode allow computer control of any of the 4 inputs (elevator {forward/back}, aileron {left/right}, rudder, collective) and rotation rate, directio and hover height. Heli-X registered version is approximately 50 euros. If you want, you can move the hover training circle, edit th StartPosition in the flying field. By default you will be standing in the hover training circle. In Heli-X, if you wander out of the hove circle, you will crash and reset at the start training position at selected elevation and orientation. This is useful for discipline.

If using Phoenix, do not use hover training mode. Unlike Heli-X, hover training does not allow unique control of each of the 4 inpu (elevator, aileron, rudder, collective). Phoenix hover training locks the collective and allows control of (elevator, aileron, elevator & aileron, elevator & aileron & rudder). Also Phoenix does not have a "hover area". I find Phoenix hover training almost worse than useless. For Phoenix, use normal flying and F3C Box for a visible area and manually crash/reset when out of area. Use Configure Settings > Physics tab > Simulation Speed slider to slow down sim when required.

After review of RealFlight 6.5 manual (and videos), RealFlight hover mode is useful, but not up to the level of Heli-X. If using RealFlight, Use hover training, medium circle, short reset delay, random upright orientation (or random inverted orientation when inverted training), control throttle, pitch, roll, yaw. You will need to adjust TX trim on the rudder to simulate constant rotation (CW CCW as required). In simulation menu, set physics difficulty to realistic. Use physics speed to slow down the sim when required.

Clearview is like Phoenix in that the autopilot in hover training mode (autopilot) does not have a boundary. At least you can set collective control for upright (but not inverted) and the autopilot will work over any background (like Nahe Sport Hall). Like phoe do not use the training mode. Use the Nahe Sport Hall as the hover environment and use the floor marking as the hover area. Us Magic Time scale to slow time.

I have not checked other simulators yet. I will add notes on them as I try them.

Whichever sim you use, ensure you know how to slow down time. You will need it when starting.

If using a sim that does not have an auto piro/rotate mode, this can be simulated with TX trim when required.

You want the sim to be able to take over nominated inputs (namely collective and rudder). You also want the sim to be able to s the hover height if collective in computer control. The hover height needs to be able to be maintained even if the heli is inverted.

When in the sim, use a 600 size heli or bigger.

Always fly Idle Up (sport mode) in the sim. Chances are you will play with flips and rolls after first training session in "play time". you are not in idle up, the motor on the sim model will die when upside down. Idle Up also requires the use of throttle hold to sto the motor. Get used to that early, you will need it.

Never use rate mode for the gyro. You will be learning backwards and sideways flying, these do not work with gyro in rate mode. is better to learn to actively steer the tail than have the heli "weather vane".

Training methodology:

Sim training sessions are to be no longer than 20 minutes. Play time after training can go as long as you like, but only after a training first. Play time after training is important to have fun and not make this so much of a grind.

Minimum rest time between training sessions is 15 mins. The rest time is important. It gives the brain time to absorb what has been learned.

You must do at least one training session each day. The repetition is important. If you skip a day, do two the next, remember mir 15min non-sim time gap between training sessions. Skip two days, to two a day for the next two days. The only way to get your flying automatic is to be doing it every day. When I started I was doing four sessions a day (two morning, two night), but I get obsessive about things like this.

The MOST important part of the training is the slow piro. Everything else (including circles, 8's and funnels) are secondary. The slow piros are the orientation correction glue that holds everything else together. I can promise you, you will get bored with the s piros. I apologise in advance for this. You do want to fly well though. Once you can slow piro and keep the heli perfectly on axis f 5 minutes, upright, inverted, CW, CCW, in wind, then you can stop doing them. Get this right and no-one will be able to match you flying. Think and the heli will do your bidding like magic.

I will ask that whatever you are weakest at is what you should practice. This is during training time only. Training time is work tin Training may be un-fun, it may be frustrating, it may be boring, but stick with it, the results are amazing.

Because training gets boring, I urge you to use a portable music player. I listen to audio-books while training the basics. You do r need your "inner voice" to fly an RC heli, same as you do not need your "inner voice" to walk. Audio input allows the mind to split attention and keep entertained while re-enforcing motor-neuron pathways that rely on primarily visual feedback. I actually look forward to the next instalment of a book while practising flying.

Lastly, set goals and rewards. This helps keep you focused. Do not cheat your goals. Don't cave and purchase early. All goals should be based on IRL progress, not sim progress. My first goal was a 1min invert hover (new heli). My second goal is a larger h I can afford to go out and get a larger heli, but will not until I reach the goal.

The progression:

This guide follows the same progression as the chadrg.com Heli Flight school method (http://www.chadrg.com/flightschool/). I gets into inverted flight quite early. Inverted hovering really is the "Alice in Wonderland" rabbit hole of RC helis. Once there, the w is never the same again. So, to quote the film The Matrix, "Buckle your seatbelt Dorothy, 'cause Kansas is going bye-bye".

- slow upright piros
- upright lazy 8s
- flips & rolls
- slow inverted piros
- upright forward circles
- upright backward circles
- inverted backward circles
- inverted forward circles
- true 8s

- funnels
- funnel 8s
- transitional 8s including funnels

Other exercises:

- 8 point slow piro hover
- compass piro hover
- piro circles
- loops
- hurricanes
- auto-rotation

Tracking hover progress:

Keep a log of:

- session date/time/length
- orientation (upright/inverted)
- direction (CW/CCW)
- sim speed
- # crashes/out of area
- comments (computer collective / computer rudder / notes on session)

The log will help you track progress. You will be surprised how quickly you will progress. Unless you track it, you will not know.

The gory details:

Lesson 1: Slow upright piros

Use the hover training mode in your simulator (except if it does not put you in a boundary like Phoenix or ClearView). You will tak complete control of the heli (throttle, pitch, aileron and rudder). Adjust the trim on your TX so that when the heli is hovering so it turning 1/4 turn every 15 seconds (1 rotation per minute).

If using Heli-X, let the sim take over collective and rudder. Set rotation speed to 1 per minute. Uncheck inverted. Set view directic to back. Set height to 2.0m. Set sim speed to 100%.

Put on some music (I use audio books), set your timer for 15 minutes and start hover training.

If you move out of your set hover area, crash/reset.

At end of session record results. Then it's play time (for as long as you want). Try some lazy 8's from lesson 2, or throw the heli round like a lunatic, by completing the training you have earned "free play"). During your play time mess with lessons 2 and 3 (an whatever else you like).

Rules for session progression:

- (Heli-X) If you have 0 crashes, remove collective from computer control for next session.
- If you have more than 7 crashes, reduce sim speed for next session by 10%.
- If you have less than 5 crashes, increase sim speed for next session by 5% (if not already 100%).
- If for 2 successive session (one CW, one CCW), you have 0 crashes (Heli-X: and collective is under your control), start less 4 next session (You should already have been playing around with lessons 2 and 3. To do lesson 4, you will need some competence in lesson 3, work on it if you need to).
- Next session must be opposite direction to current session (ie. if current CW, next must be CCW)
- Must have a 15 minute break (at a minimum) from the sim before starting a training session.

The constant rotation means you cannot cheat. All recoveries must be in the direction you are currently facing.

The purpose is to eventually not have to think about cyclic corrections. Initially you may need mental prompts such as:

- When nose in, prop up direction that is falling (prop).
- Nose left or right, push stick toward nose will bring heli to you (tail-away).

Eventually you will not need them.

If you are comfortable doing this and you only want to fly scale (no inverted), feel free to move onto "Lesson 6: Upright forward circles". I would highly recommend working on "Lesson 2: Upright lazy 8s" in play time till no crashes when slow piro upright hovering. The other thing to remember is that scale also includes 8's and nose-in circles (funnels), so upright true 8's from "Less 10: True 8s" and upright nose-in funnels from "Lesson 11: Funnels" would be on the curriculum.

Lesson 1 - Slow Upright Piros - From tail-in to all 8s and funnels in 6 months (2 min 12 sec)



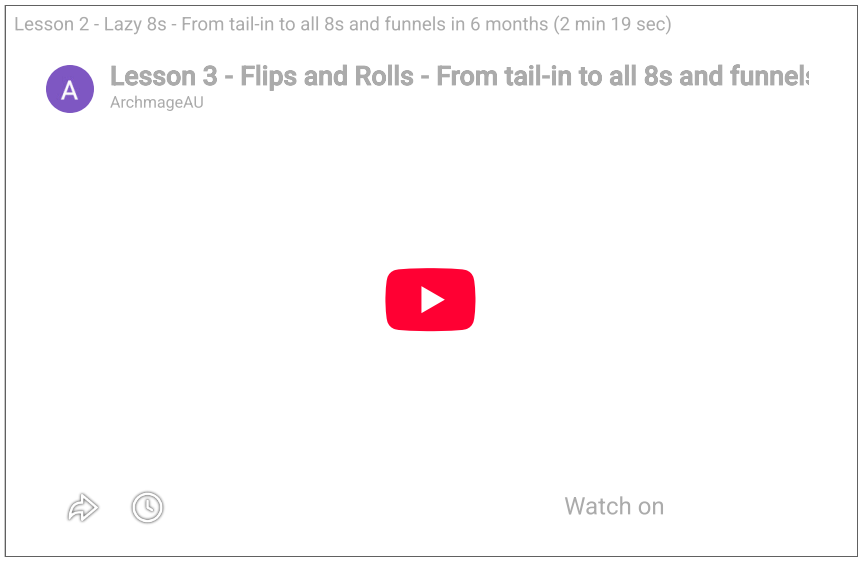
Lesson 2: Upright lazy 8s

Start trying these in play time. Centre towards you at this stage.

A lazy 8 is a left/right then right/left pass with an optional turn at either end. See: Flat Figure of Eight (http://www.rchelicopterweb.com/Learn...uraOfEight.htm)

You can work this up to a nose leading 8. See: Flat figure of eight with the nose following flight direction (http://www.rchelicopterweb.com/Learn...tDirection.htm)

As you progress with these, the turns at the ends will get better and better.



Lesson 3: Flips and Rolls

Let the fun begin. Also start trying these in play time.

Try front flips, back flips and side rolls.

For a flip, from a level hover (tail-in), start to climb with some collective. Push and hold a cyclic in the direction you want to flip, when the rotor disc is vertical, switch from positive to negative collective. Initially, keep it rolling and switch from negative to positive collective when the rotor is vertical again. Stop when level upright again.

The aim for the flip or roll is to do it losing minimal height and travelling the minimal distance. Try do it within a training square c circle, consistently.

Once comfortable with doing full flips and not losing height, try stop a flip halfway (inverted), then continue and flip back to upri Slowly increase the time inverted. Always have a way to flip back to upright.

Normally, you will find one flip direction comfortable than any other. For me it was front flip. Once you have a comfortable flip, ignore that one and practice all the others.

You can do funky stuff with flips and rolls. I like to start a back flip and stop vertical, pull negative collective the swing the heli through flat inverted to vertical on the other side, then switch to positive collective and swing back down to upright. Another I en is doing the same but sideways. Edge, swing, edge, swing.

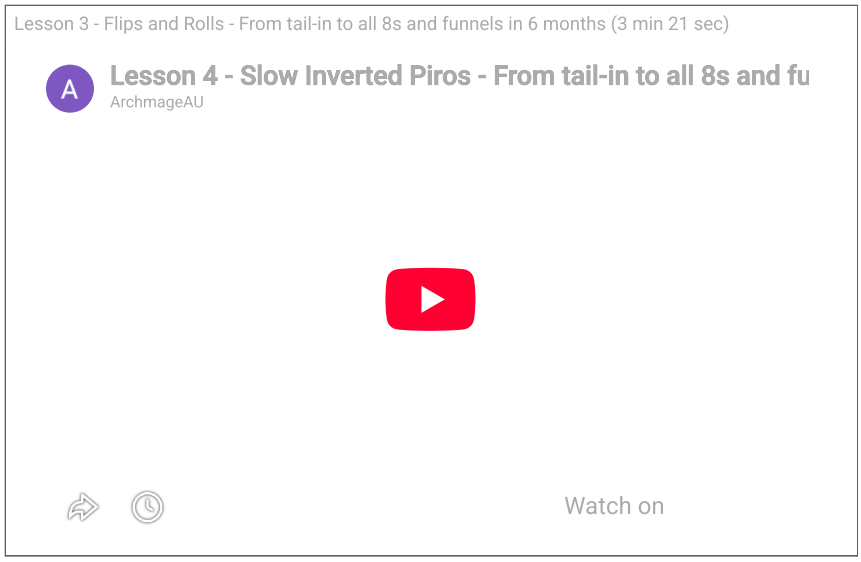
Once you have flips and rolls going, you could try a rainbow. From stable upright hover, push hard positive collective and pull ger back on the elevator. When at an inverted 45 degree angle, release the elevator and and pull hard negative collective. The heli should almost stop in the air, then start back in the direction it came from. Gently push the elevator forward and the nose will cre the rainbow in the direction it came from. When at upright 45 degree angle, release the elevator and push hard positive collectiv (rinse/repeat or level off). This is the basis to tic-tocs. Hard collective, softer elevator. As you get better with orientations, rainbo become more consistent and you will be able to do corrections in the middle of the manoeuvre.

There are almost unlimited flip and roll based tricks to try:

- Diagonal flips, start with tail at 45 degrees, and flip using both elevator and aileron.
- 1/2 flip and 1/2 roll sequences, ie. flip nose-in inverted, roll to nose-in upright, flip to tail in inverted, roll to tail-in upright.
- Once comfortable with inverted hovering, try start flip or roll from inverted and finish inverted.
- Rainbow with a stop half way (mid stick collective in the centre)
- Rainbow with a half roll at the midpoint.
- Rainbow with a half piro at the midpoint.

The point here is to experiment and play. You have the safety of the sim, so find out what works for you and looks "cool".

Flips and rolls will come important in the next exercise. Slow inverted piros.



Lesson 4: Slow inverted piros

"We're not in Kansas any more Toto!"

When you start this you will HATE ME. When you "get it" you will feel like you have just won an Olympic gold medal.

We have not been recording lessons 2 and 3 in our log. We should still be doing Lesson 1 daily, however, now we will start invert

Use the hover training mode in your simulator (except if it does not put you in a boundary like Phoenix or ClearView). You will tak complete control of the heli (throttle, pitch, aileron and rudder). Adjust the trim on your TX so that when the heli is hovering so it turning 1/4 turn every 15 seconds (1 rotation per minute).

If using Heli-X, let the sim take over collective and rudder. Set rotation speed to 1 per minute. Check inverted. Set view direction front (or back). Set height to 2.0m. Set sim speed to 100%.

Put on some music (I use audio books), set your timer for 15 minutes and start hover training. This time you may need to flip or to start with. If you wander out of area on the flip or roll, that is a crash. Start again. (see why you wanted to get these flips accu within the box or training area 😀)

If you move out of your set hover area, crash/reset. If that means you need to re-flip or roll, then do it.

At end of session record results. Then it's play time (for as long as you want). Try anything you feel like (loops, tic-toc's, rainbows stall turns, circles).

Rules for session progression are the same as upright, setting modifications only for inverted:

- (Heli-X) If you have 0 crashes, remove collective from computer control for next session.

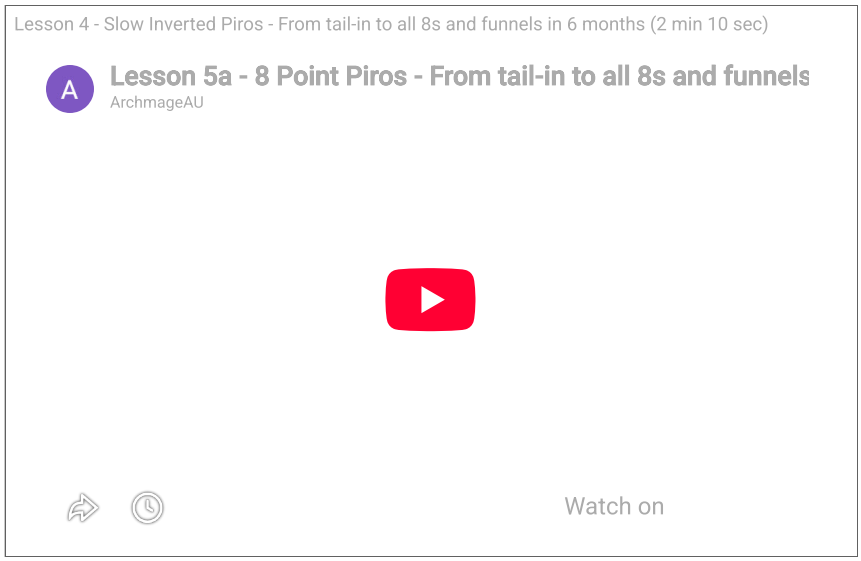
- If you have more than 7 crashes, reduce sim speed for next session by 10%.
- If you have more than 50 crashes, reduce sim speed for next session to 50% (or lowest possible sim speed).
- If you have less than 5 crashes, increase sim speed for next session by 5% (if not already 100%).
- If for 2 successive session (one CW, one CCW), you have 0 crashes (Heli-X: and collective is under your control), start less 5 next session.
- Next session must be next in the following 6 orientation list: invert CW, invert CCW, upright CW, invert CW, invert CCW, upri CCW (this way we interleave upright with inverted practice. (Upright is always at full speed, inverted is at the previously determined inverted speed).
- Must have a 15 minute break (at a minimum) from the sim before starting a training session.

The constant rotation means you cannot cheat. All recoveries must be in the direction you are currently facing.

The purpose is to eventually not have to think about cyclic corrections

The first time you do this you are likely to crash more than 50 times in 15 minutes. That's OK. You are getting practice at tight fli and starting to learn. The first time you get a complete rotation you will feel great. When to get to 100% speed you will feel awesome. When to get 2 successive 0 crash sessions, reward yourself, you have passed the hardest lesson.

During this training, play time is the most important. This is the hardest laborious and frustrating.



Lesson 5a: 8 point piros

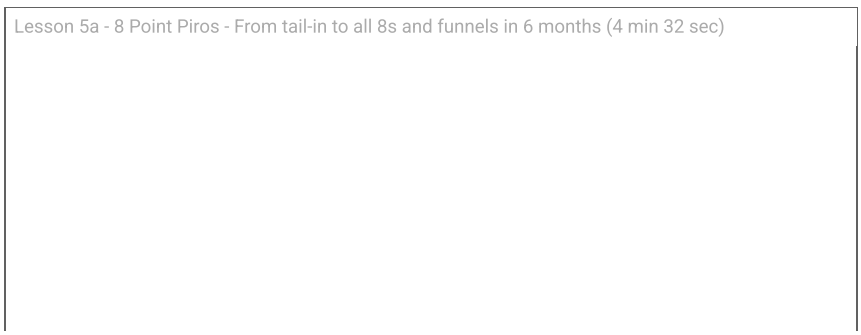
Now you are a novice hoverer. You can hover full speed upright and inverted. We can shelve hovers forever right? WRONG. You want to be amazing at this.

Because you can recover in any orientation at full speed upright and inverted, time to try in real life. Take your CP heli out to a lar field, take it at least 3 meters high and flip and hold. Any issues, flip back to upright. All you are doing at this stage is hover. You have not had control of rudder yet in your training, so likely any inverted rudder correction will be incorrect. Fortunately you can recover no matter the orientation 😀 (You've done it for 2 straight 15min periods in the sim with constantly changing orientation, should be easy).

The hover training will continue (once a day). One day inverted, one day upright. The difference this time is no rudder control. Yo are to doing the rudder and you will do 8 point piros. Stop at each point for at least 1s (N, NE, E, SE, S, SW, W, NW). Each session must be 10 rotations (5 CW, 5 CCW). Record results in your log.

Once you can do these without issue 10 days running (5 upright, 5 inverted). Move onto Lesson 5b

(this is where I am up to after 6 months)

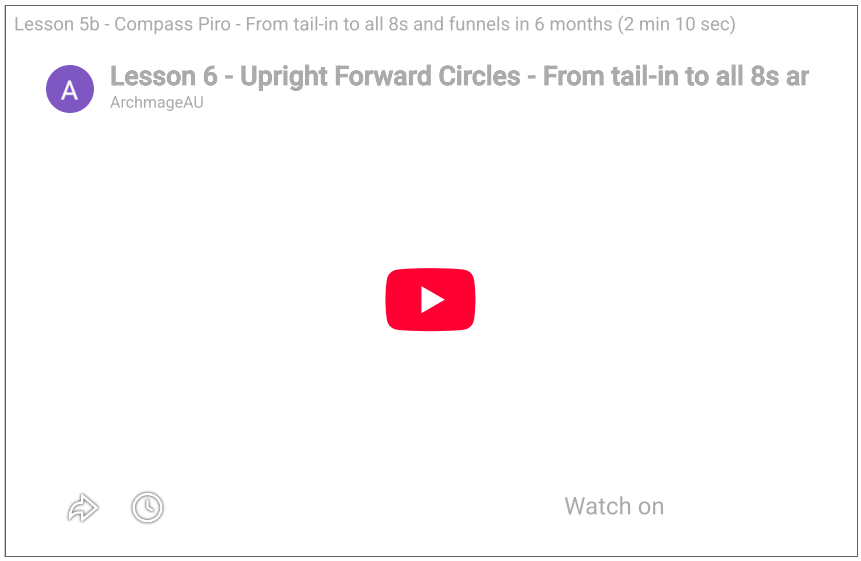




Lesson 5b: compass circles

Now you are a intermediate hoverer. You can hover full speed upright and inverted with tail control. We are done right? WRONG.

A compass circle is like an 8 point piro, but you precise hover to each compass point (N, NE, E, SE, S, SW, W, NW), nose toward th centre, move to the centre, then back out to the compass point, then hover, nose toward the centre of the circle, to the next compass point. Do this once each CW and CCW, upright and inverted.



Lesson 5c: Other hover exercises

Now you are an advanced hoverer. You can hover full speed upright and inverted with tail control, 8 point piro and compass circl We are done right? ALMOST (well not really, but I need to give you hope)

Each of our hovers since the slow piros has been stilted (non-continuous). Time to speed things up, but slowly. Remaining exercises are:

- Travelling slow piros (follow lines on the ground (circles, squares) while slow piroing)
- Travelling fast piros (follow lines on the ground (circles, squares) while fast piroing)
- (do all the previous hover exercises, but use wind 15-30 km/h, then gusty)

Once you can do all this, you are probably flying on the deck smack with 700s and officially an RC heli god.

Unfortunately no video for this one. The author is not an RC heli god yet 😀 (as you would see from the other videos).

Lesson 6: Upright forward circles

Now we move onto the slow upright forward circles.

You can hover in all orientations, so this should be simple 😀

There are two ways to approach this. Continuous banked circles or hover to points. We are going to start continuous banked circles. CW and CCW.

We are not flying around us (as I have seen some suggest), we can hover if we get stuck 😀 .

Start at tail-in hover and start travelling away from yourself and steer a direction with the rudder and the the same direction with aileron. This is a coordinated turn (rudder and aileron go same way). As the turn starts, pull back slightly on the elevator. We only want a shallow bank turn.

Now the fun begins. We need to balance rudder, collective, elevator and aileron to make the turn. Yes, ALL OF THEM.

As we are using a HH tail gyro (remember, no rate mode), so you must ALWAYS steer the rudder around the turn.

Next is the aileron, tighter into the turn you will slip and lose height (and possibly speed up). Tip out of the turn and you will gain height (and possibly slow down). This is your bail out. Level the disk and pull back more on the elevator.

Next is elevator; less elevator, wider circle; more elevator, tighter circle.

Last is the collective. More collective, faster turns, less collective, slower turns.

What you are aiming for is a consistent circle.

When turning ALWAYS keep the rudder moving. Most common beginner mistake is not keeping the tail turning. This will save yo more often that anything else. Next most common mistake is too much bank angle, the easiest way to avoid this is rock the aile back and forward till you get a feel for the angle required for the turn rate and speed.

Once you get these two, feather the elevator (pitch) to keep the turn happening (not too much you spiral in and stop, not too little you lose the circle). Lastly, use collective to moderate the speed.

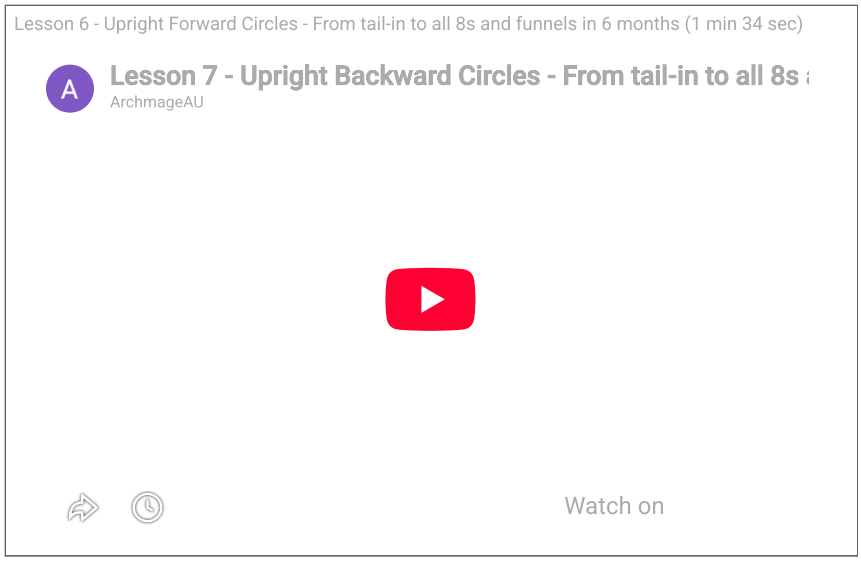
The way I first got these was to think of flying the disc. Assuming the tail is just an indicator for control inputs, what way do you need to tilt the disc to maintain a circle. Imagine a control stick out the top of the rotor disc. If you pull the stick back toward the tail, the front of the disc lifts (this is your elevator). If you push the stick left, the right side of the disc lifts (this is your aileron) { mode 2 they are on the same stick. Yay!! 😀 }. If you keep the tail level round the circle (horizontal to the ground) using rudder, the only input you need is pulling back (elevator). If it starts to slip into the circle, steer out with your aileron. For more information o "flying the disc", see the end of this guide near the references

CW and CCW feel different, you will need to practice both.

You want to get 5 of each circle (CW and CCW).

Once you can do 3 consecutive or these in each direction without stopping or wandering, time to try backwards.

Are you remembering to practice your slow piros or 8 point piros?



Lesson 7: Upright backward circles

I've read someone state that those who can fly backwards circles and 8's must have sold their soul. Maybe not sold, but at least second mortgaged 😀

This is WHY we are using a HH tail gyro (this will not work in no rate mode).

The first time you try do this as a circle you are likely to crash. It will feel so unnatural. Persist and you will get there.

A backward circle is the same as an upright forward circle, just backwards. Simple hey!

A backward circle is an uncoordinated turn. This means the rudder goes one way, and the aileron goes the other. You must push forward on the elevator to go round the turn. The temptation is to over-steer these. Don't! You want nice large even circles.

Yet again, rock the aileron to find the groove for the circle. Like the forward circle, you must constantly use the rudder to keep th circle going.

If you have get in trouble, pull the rudder harder into the circle, especially if you have tipped the aileron to far into the circle and a staring to drop.

With practice you will get to feel how hard to "push off" and how hard to push the elevator to keep the circle going without accelerating or stopping.

The hardest thing I find with backward circles is keeping the tail parallel to the ground. Once I get it right, the circle just seems tc flow.

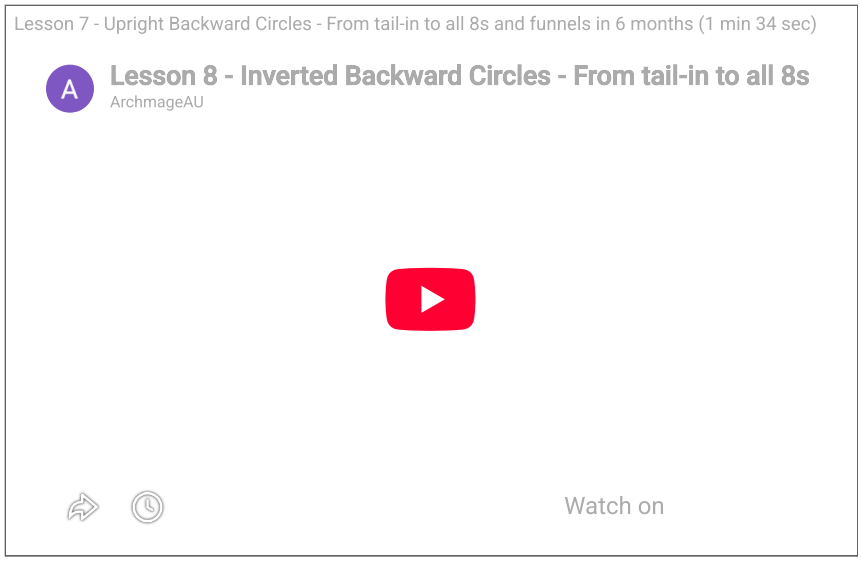
If needing to bail-out, level the disc and add collective. Combine this with the push of the elevator that keeps the circle going, yo will stop quickly.

You want to get 5 of each circle (CW and CCW).

Once you can do 3 consecutive or these in each direction without stopping or wandering, time to try inverted.

Are you remembering to practice your slow piro's or 8 point piros?

Slow or 8 point piros anyone?



Lesson 8: Inverted backward circles

Oddly enough, I find inverted backward circles easier than inverted forward circles.

I find the easiest way to get into these is to front flip from forward flight and be a touch late on the negative collective and let the flip continue to give the push-off.

The inverted backward circles are similar to forward upright circles, only collective is reversed and you steer to opposite way to normal (ie. turn left to go right). Inverted backward circles are coordinated, which means the rudder and aileron move in the sam direction.

Keep your circles higher than your upright circles. If you judge heli height by centre of mass, your blades will be closer to the ground. This means you will blade strike easier. So keep em' high.

The other thing you will find is that the heli will slip sideways through the air more often if out of balance. This seems to be beca the thrust of the blades is from below the centre of gravity instead of above it like upright. To prevent this, you will need to use th aileron to tip the centre of mass further into the circle, then level out a bit more to maintain the circle diameter. All the time steer with the rudder (and maintaining height, and maintaining speed). Simple

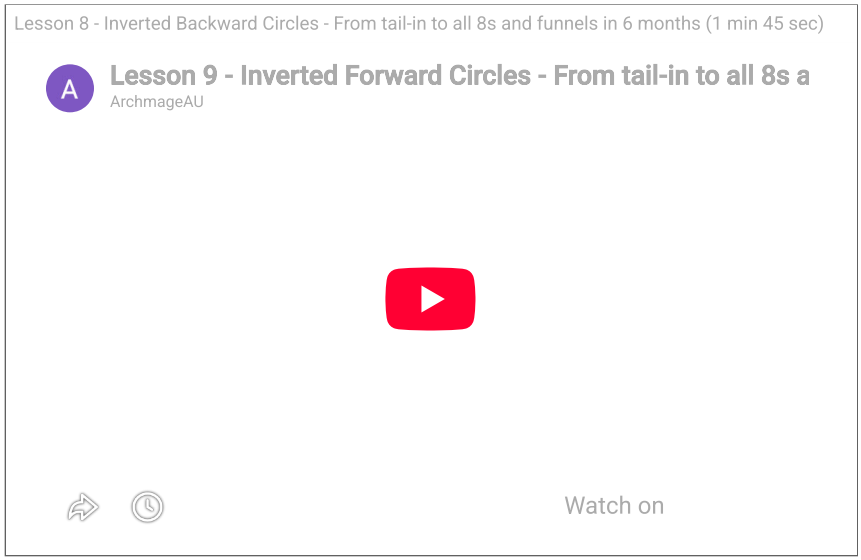
Remember to practice your bail-out. Initially your bail-out disc level will be wrong (as you are trained in upright circles). This is wl

the sim practice is for. If you rock the aileron, even slightly, to groove the circle, you are less likely to level the wrong way. Other b out is to pull harder on back elevator and full negative collective. This will buy you height to sort out what to do. Because you are expert upright hoverer, you should have no problems from there.

You want to get 5 of each circle (CW and CCW).

Once you can do 3 consecutive or these in each direction without stopping or wandering, time to try inverted forward.

* Do you know what I am going to ask? *



Lesson 9: Inverted forward circles

Almost there. The more I practice this, the more comfortable this becomes. Still my hardest, but not by much.

I find the easiest way to get into these is to roll in from forward flight. My other usual way of starting these is a front flip to invert hover, piro 180 degrees and then push off.

The inverted forward circles are similar to upright backward circles, only collective is reversed and you steer to opposite way to normal (ie. turn left to go right). Being an uncoordinated turn, the aileron goes the opposite way to the rudder. Assume the rudde sets direction and the aileron is used to balance the turn.

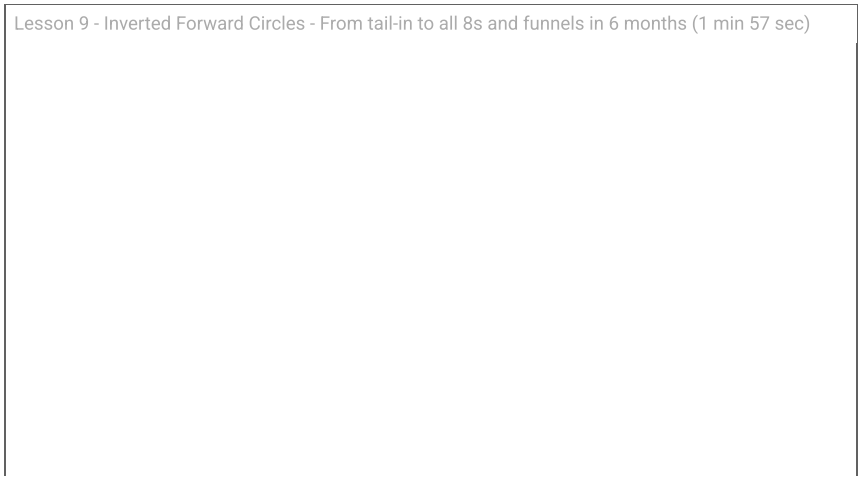
Heli slip on inverted forward circles is similar inverted backward circles. You still need the same juggling act that you learned fo inverted backward circles, but with the aileron reversed. How hard can that be 😀

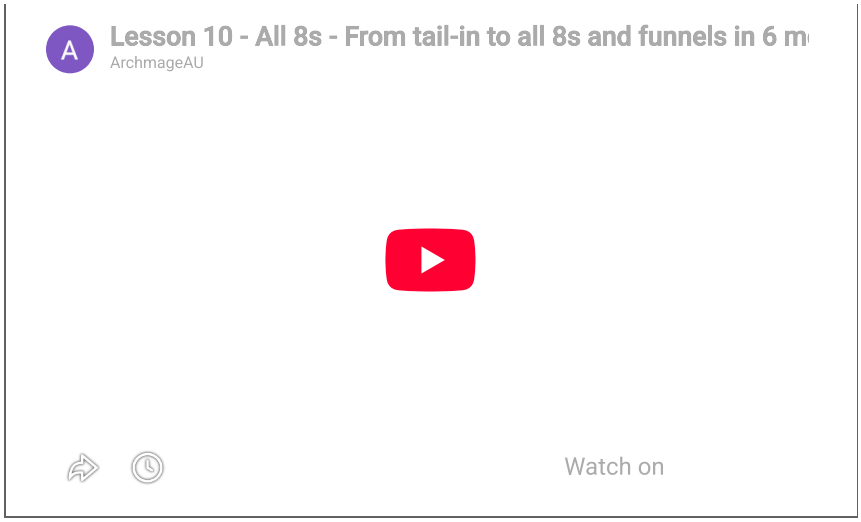
Just like the inverted backward circles, remember to practice your bail-out. Best is level disc and push elevator to stop, but the d collective, push elevator (to gain height, then recover also works). The only problem with the collective, elevator bail-out is that ir when you get to hurricanes (really fast, high angle circles), this move may put you in more trouble.

You want to get 5 of each circle (CW and CCW).

Once you can do 3 consecutive or these in each direction without stopping or wandering, time to put it all together with true 8s.

* You know I will keep nagging you about the slow or 8 point piros *





Lesson 10: True 8s

This will be your first real foray into transitions. It's only a simple transition, but one none the less.

Now that you can fly circles where-ever you want, time to put aside the lazy 8's forever and start doing some real flying.

The goal with 8s is to make them even. You will need to control the size, shape and speed of your circles. It will feel like just as soon as you have the groove in one circle, you will be changing to another other one.

I find it easier to start with outward 8's. This is an 8 where at the centre of the 8 the heli is travelling away from you. This is the opposite to the starting lazy 8's. The advantage of this is that the heli is not coming straight at you while switching from one circ to another.

As well as controlling the circles to make sure they are even and the transition is at the centre, you want to stop the heli climbinç through the centre. When you straighten up out of a circle, the tendency is to climb unless you back off the elevator. Through practice, you will learn how to much to back off to keep the transitions level.

It is tempting to short-cut the centre transition. Don't! Complete the first circle before starting the next.

Once you can do outward upright forward 8's, progress to inward upright forward 8's.

The full progression sequence will be:

- outward upright forward 8's
- inward upright forward 8's
- outward upright backward 8's
- inward upright backward 8's
- outward inverted backward 8's
- inward inverted backward 8's

Once you can do 2 of one without stopping, proceed to the next.

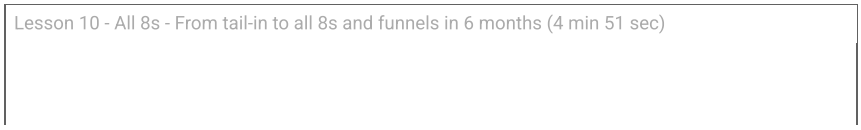
Eventually, in one training session you should be able to do 5 of each (That's 80 circles in 20 minutes or less, or less than 15s pe circle as you will also have transitions between 8s).

Before progressing to funnels, be sure you can do 2 of each in a single session without crashing, bailing out or leaving area.

What my practice session was like when at this level of training was:

- Inverted 8 point piro CW & CCW.
- Upright 8 point piro CW & CCW.
- 5 of each 8 (outward/inward, upright/inverted, forward/backward).
- Inverted 8 point piro CW & CCW.
- Upright 8 point piro CW & CCW.
- Play time.

The 8's were identifiable, but pretty messy. It took me until month 3 to get to this.





Lesson 11: Funnels

By the time you get here, you will almost be able to do circles in your sleep (This is the idea).

By this time you will have experienced messy circles, recovered circles in all orientations, blown out circles, blown up circles, slipped and crashed circles and probably everything in between.

Now we are going to try and fly some deliberately messed up circles as opposed to the accidentally messed up ones.

Funnels relate to circles:

- Upright nose-in funnels feel like upright forward circles with the tail swung out.
- Upright tail-in funnels feel like upright backward circles with the tail swung in.
- Inverted tail-in funnels feel like inverted backward circles with the tail swung in.
- Inverted nose-in funnels feel like inverted forward circles with the tail swung out.

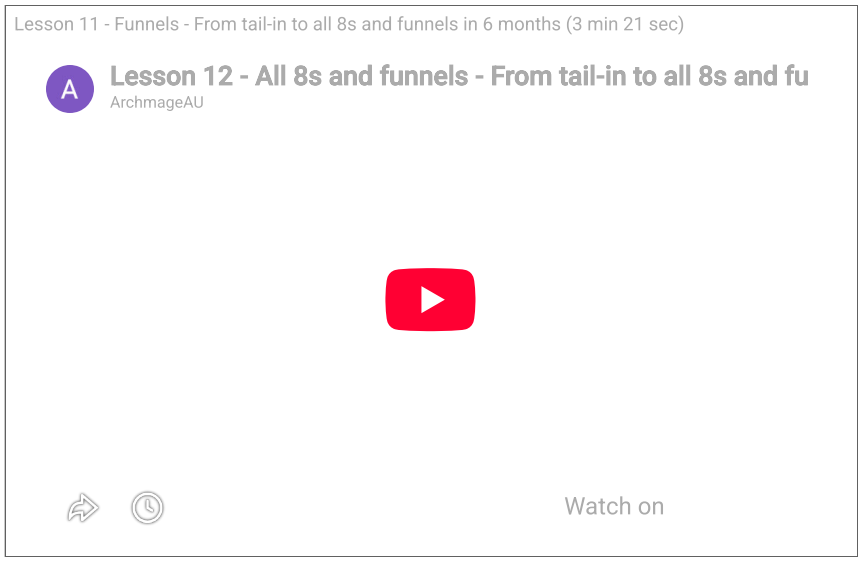
Pick the circle you are most comfortable in. Start to fly the circle. Now pull (or push) the rudder to slide the circle into it's relevan funnel orientation (ie. If doing a forward CCW circle, pull the rudder left a little more to slide the tail up and out of the circle, but keep the circle going with the aileron. What you will find is that the aileron is working like the elevator, maintaining the circle spe and the elevator is working like the aileron, maintaining the heli bank in the circle.)

This will take a while to get used to (just like circles initially). The hard part will be controlling the funnel size and shape. Remem to rock the elevator (instead of the aileron) to keep the angle of the funnel and try and groove the path.

The main pint that makes a funnel good is the angle of the tail. Try get the tail to point directly at (or away from) the centre of th circle. Remember to steer the funnel using the rudder (same as a circle).

You want to be able to do all of these (5 or each) before progressing to funnel 8s.

* ? *



Lesson 11a: Funnel 8s

If you get to here, you have completed the original purpose of the guide.

You should now be able to fly any of the circles or 8's safely with your actual CP heli (even in the wind). You may also be starting fly funnels with you actual CP heli. We are not finished yet though.

Funnel 8's is an exercise I developed just to help me achieve a personal goal of efficiently flying all 8 and funnels to complete in single 5 min flight. I have yet to achieve this, but I am close.

Most funnel 8s you will see are high angle funnels, tipping from upright to inverted funnels, often tail down. These look impressi What this exercise is for slow funnels an transitioning from tail-in to tail-out, while remaining upright or inverted.

The advantage of this is that these funnel transitions can be practice SLOW.

Instead of forwards or backwards, either the left side of the heli leads, or the right side. An outward upright right funnel 8 consis of a upright nose-in CCW funnel followed by an upright tail-in CW funnel.

Consequently there are 8 funnel 8's to practice.

- outward upright right funnel 8
- inward upright right funnel 8
- outward upright left funnel 8
- inward upright left funnel 8
- outward inverted right funnel 8
- inward inverted right funnel 8
- outward inverted left funnel 8
- inward inverted left funnel 8

Once you can do these consistently, move on to Transitional 8s including funnels.

I did not make a video for this as it covered in lesson 12 which incorporates the funnel 8's between the normal 8's.

If you are really adventurous, you can morph the funnel 8's from upright to inverted. In this you effectively flip the heli from uprig to inverted through the change and the tail will stay in it's current orientation (in/down or out/up). These are best done if your funnels have a rotor tilt greater than 45 degrees. As you are learning, your funnels will tend to be at a lesser angle though.

Lesson 12: Transitional 8s including funnels
This is the last exercise in the guide. This is the hardest exercise I use. It takes me 20 minutes to do 3 of these (bookended by inverted and upright, CW and CCW 8 point piros).

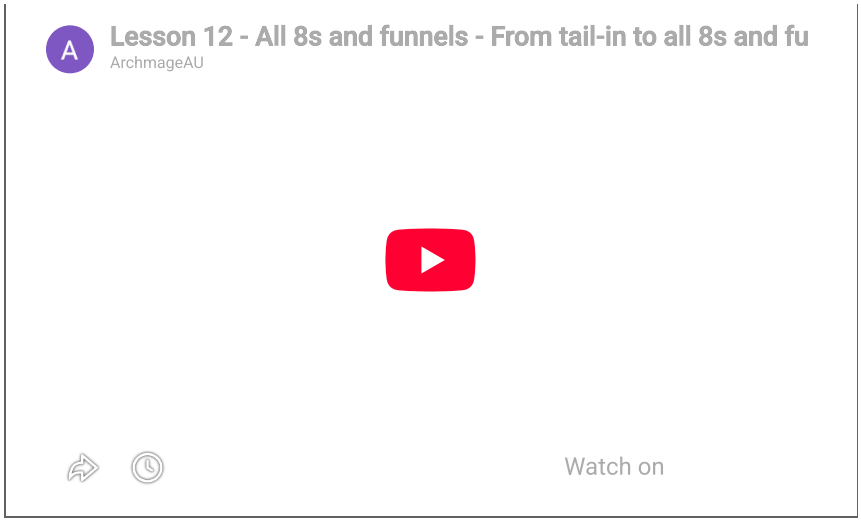
The sequence flows together and does 2 of each circle. The point of this exercise it to ensure all the weak points of your flying a tested. Once you can do this consistently, you should be able to transition smoothly between most flying elements.

The sequence is:

- outwards upright forwards 8.
- outwards upright nose-in to tail-in funnel 8.
- outwards upright backwards 8.
- outwards upright tail-in no nose-in funnel 8.
- - (flip)
- outwards inverted backwards 8.
- outwards inverted tail-in no nose-in funnel 8.
- outwards inverted forwards 8.
- outwards inverted nose-in to tail-in funnel 8.
- - (traverse to opposite 8)
- inwards inverted backwards 8.
- inwards inverted tail-in no nose-in funnel 8.
- inwards inverted forwards 8.
- inwards inverted nose-in to tail-in funnel 8.
- - (flip)
- inwards upright forwards 8.
- inwards upright nose-in to tail-in funnel 8.
- inwards upright backwards 8.
- inwards upright tail-in no nose-in funnel 8.
- - (traverse to opposite 8)

Complete this sequence in a single real flight and you have done something extraordinary.

Lesson 12 - All 8s and funnels - From tail-in to all 8s and funnels in 6 months (5 min 13 sec)



From here, the sky is the limit.

If progressing to 3D, the main skills beyond this you will want to learn are piro moves. Just by being able to do these, you should able to to piro circles and maybe piro 8s. Perfecting piro flying is a whole new topic (and my next adventure in RC heli flight).

Flying the disc:

This is a note on a technique commonly referred to as "flying the disc". What it means is that you no longer think of the heli as ai aircraft, but a flying disc that you control. It's a bit of mental gymnastics, but once understood it can change the way you think of heli flying forever.

First thing to realise is that you may not always be able to see the disc. Black blades become invisible at speed, even day-glow orange or reflective can be hard to see at times. The position of the disc can always inferred by the fuselage and the tail when nc seen. Therefore it is possible to fly the disc even when it is effectively invisible (warned you about mental gymnastics).

For the purpose of this section, assume you can "see" the disc.

Flying a heli is often described as balancing a marble on a plate. To move the marble you need to tilt the plate. Once the marble starts to move, you need to move the plate under the marble to keep it balanced. if the marble is too unbalanced on the plate, it falls off the edge. It is simple to see how to move the marble from place to place by tilting and levelling the plate. "Flying the disc tilting this plate but with a twist.

Picture this plate that is for balancing the marble has a virtual "ghost stick" protruding up from the middle of it. If you push this "ghost stick", the plate tilts (and the marble) moves. If you "bump" this "ghost stick" forward, the plate tilts momentarily, the marb starts to move forward, so the plate moves with the marble and continues to move till the "ghost stick" backward. Now imagine, there is a tail-in ghost fuselage hanging below the plate and the "ghost stick". This "ghost stick" is your mode 2 cyclic stick (eleva - forward/back, aileron - left right). Finally image we put a compass knob on this ghost stick labelled with N, E, S & W, N points towards the heli's nose, S towards the heli's tail, E right, W left.

From our previous example, if we bump the compass knob on the top of the ghost stick to the N, the heli will move forward. Bumping the compass knob to the S will stop it.

Now turn the whole thing 90 degrees to the the left. W is now facing toward you. To push the heli away, bump E, to stop bump W

Now lets really stress the grey matter. Assume our arrangement is in a inverted nose down funnel. Skids are facing toward the middle of the circle. Funnel is going to go CCW. Which way do you need to push the ghost stick to keep the funnel going? Pushir N will tip the nose up and flatten you out of the funnel. Pushing S will point the nose toward the ground and make the funnel slip down. Mentally flip the heli over and you will see W is the direction of travel. So you need to push W to keep an inverted nose-in CCW funnel going.

So all we are doing "flying the disc" is pushing this stick around to tilt the disc in the direction we need it to go. The collective do the thrust, the rudder does the twist and the stick does the rest.

So all we need to direct a heli is know where the tail is and which way the heli is headed (we can infer the disc from that) and ste appropriately. We can figure out controls for a move and just do it. Eventually the brain gets out of the way and the fingers/thum work out what to do. A state of "Unconscious Competence".

For another perspective on this see gitbse's thread "Ok, here goes. With pictures!" (https://www.helifreak.com/showthread.php?t=530707)

References:

- HeliFreak thread "Just confirming I am on the right track" (https://www.helifreak.com/showthread.php?t=493289)
- chadrg.com Heli Flight school method (http://www.chadrg.com/flightschool/)
- RADD's school of rotary flight (RaddsFieldManual.pdf - attached now as the dream-models site where the original http://www.dream-models.com/files/docs/FieldManual.pdf was housed is now offline)
- 1st U.S. R/C Flight School - Transmitter Handling (http://www.rcflightschool.com/TransmitterHandling.pdf)
- F3C / FAI stuff for precision hovering (http://www.f3c-heli.de/downloads/cat...reibungen.html)
- Training videos from Caspart (http://www.rc-heli.de/board/showthread.php?t=79801)
- Flat Figure of Eight (http://www.rchelicopterweb.com/Learn...uraOfEight.htm)
- Flat figure of eight with the nose following flight direction (http://www.rchelicopterweb.com/Learn...tDirection.htm)
- HeliFreak thread "Ok, here goes. With pictures!" (https://www.helifreak.com/showthread.php?t=530707)

Addendum:

For those wanting to track progress and fine more things to progress with, I can recommend: http://rcheli.club/moves/

This site can help you track proficiency and suggest more maneuvers to keep flying interesting and progress beyond this course

Attached Files

RaddsFieldManual.pdf (175.1 KB, 4283 views)

psawya, mystar60, bbow01 and 22 others like this.

_____

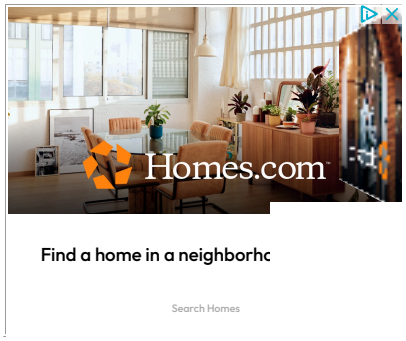
TRex 700E Pro DFC HV (BD3SX) - Gaui X5 (6S/BD3SX), Blade 130X, RealFlight, Heli-X, Taranis+DSMX

Last edited by ArchmageAU; 06-19-2017 at 03:06 AM.. Reason: Fixed Youtube URLs

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05-19-2013, 12:10 PM #2 (perma

Stubbz
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Start with stopping when the disc is flat. When you are getting better, you can stop the helicopter earlier, for example at a 45 degrees angle. If you do them very agressive, you will be almost vertical at each end.

Please note that to stop the helicopter, you'll give pitch to stop it, and then go back to the pitch you need to hover at that point. H much pitch is needed to stop the helicopter is basically trial and error, but giving to much will just make you go up a bit, where nc enough pitch might make you go to low, so start on the high side to be safe.

Try to keep your rainbows round. So in the middle of the rainbow, the disc should be exactly vertical. This will help with more advanced tricks later. Of course, very low agressive rainbows will be almost vertical for the entire lenght of the move, so it won't round. However, making it round will help in the beginning to prevent the helicopter from overshooting at the stop.

Some fun versions:
Give mid stick on the collective in the middle of the rainbow(where did we do that before?) and then continue your rainbow. This looks awsome.
Another fun one, try going mid stick, then turn the helicopter with aileron 180 degrees and continue. The orientation changes no This is hard to do smooth, but also looks very nice.
Another fun thing to do would be throwing half a piro in the middle. To be able to do this, you need to be able to do the rainbow i both orientations.
I can go on and on, but the point is that you can do tons of things to make it more interesting.

Archmage, I think it is awesome you do this! You are proof that focusing on the basics first will make learning to fly a lot faster. Keep up the good work!

Sent from my iPad using Tapatalk HD

Dysl3xicDog likes this.

_____
Gaui X3, Gaui X5, Gaui X7, Heli-X, Futaba 14SG

Profile Posts Quote Quick R

05-19-2013, 08:58 PM #5 (perma

ArchmageAU
HF Support



Thread Starter

My HF Map location
Join Date: Jan 2013
Location: Melbourne, Australia

blade450, gothicbunny,

Thank you for the feedback. I plan to keep revising this for a while (till I can no longer edit it). I will incorporate your feedback.

Just a note: Although I am doing funnels after 6 months, I appreciate not everyone advances at this rate. Some will go faster, so slower. At no point do I state the reader must progress at a particular pace. What I do state is that you should complete certain skills to a level of measurable competence before continuing (however short or long it takes).

I also do not specify the size of the CP heli for in real life work. This will work if you CP heli is a TRex 1000 or a nano (or anything between). I just stated that *I* used Blade micros (and why).

Quote:

> What you will need:
> ...
> - **A collective pitch heli** (To practice IRL what you learn. I used Blade micros (nano and 130x) because they are resilient (crash without breaking), and spares are quick and easy to obtain).
> **- For more fun (completely optional)**
> * A micro 3ch or 4ch coax heli (like a syma s107 or a Blade mCX2 - I have a large syma s033 and a micro metal 3ch)
> * A micro FP heli (like a Blade mSR or Blade mSRX - I have a "Mini Twister Pro sport - mode 2" which takes nano batteries).
> - Note: These helis will demonstrate how far you have come and are fun to let other try. I also have a "tame" setting on my transmitter for the nano, so I can let other complete novices try them without fear of wrecking them. ...

The times between training sessions and repetition schedule of training sessions suggested is optimum for an "average" persor Skill acquisition progresses with consistent, continued repetition. Short training every day or two lead to faster skill mastery thar larger training sessions once a week. I do not have a specific reference for this information, but I am sure a brief search would fi it (and it supports my personal observation in both learning and watching others learn complex skills).

_____
TRex 700E Pro DFC HV (BD3SX) - Gaui X5 (6S/BD3SX), Blade 130X, RealFlight, Heli-X, Taranis+DSMX

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05-19-2013, 09:37 PM #6 (perma

chris-crash
Registered Users



Fantastic thread! Thanks for taking the time to post this!